

VINAYAK TYAGI

Jaipur, India Email: vinayaktyagi.ed@gmail.com LinkedIn: linkedin.com/in/vinayaktyagi10
GitHub: github.com/vinayaktyagi10

EDUCATION

Manipal University Jaipur

B.Tech, Computer Science & Engineering | CGPA: 8.06

Jaipur, India

2024 – 2028

EXPERIENCE

DevOps Intern

Software Development Centre, MUJ

Aug 2025 – Present

Jaipur, India

- Identified and patched **CVE-2025-55182** (CVSS 10.0, critical — unauthenticated RCE in React Server Components) across **8 production services** within **24 hours** of disclosure; hardened all Docker images to non-root execution during the same rollout.
- Restructured Dockerfiles and authored incremental **GitHub Actions** build workflows across **45 repositories** — reduced image sizes by up to **82%** (2 GB → 358 MB) and **79%** (786 MB → 164 MB), cutting registry storage and deployment times across the CI/CD fleet.
- Deployed **Tailscale** Zero Trust mesh across Hostinger and bare-metal nodes, eliminating all public SSH exposure; enforced kernel-level resource governance via **cgroups** across the containerized production fleet to prevent resource contention across co-located services.
- Maintain deployment infrastructure for **3 production academic platforms** (NPTEL certificate verification, Elective Management System, IT ticketing) serving **2,160 CSE students** at Manipal University Jaipur across 45 repositories.

RECOGNITION

Cited at SANSFire 2026 — “*Skynet 1.0*”

July 2026

Dr. Johannes Ullrich, Dean of Research & Director, SANS Internet Storm Center

- MIRAGE**'s dataset was cited in Dr. Ullrich's SANSFire 2026 keynote — he is the founder of DShield.org and one of the most recognized figures in global network security.

PROJECTS

MIRAGE | Deceptive SSH Honeypot & Threat-Intelligence Platform

Live

Go • PostgreSQL • Python • Docker Compose • GitHub Actions • Prometheus/Grafana

Dataset DOI

2-person team

- Built a production SSH honeypot from scratch in **Go**: custom protocol handshake, credential-auth logging, and a byte-by-byte interactive fake shell (`ls/cd/cat`, pipes, redirects, command chaining) with persisted working-directory state per session.
- Designed and shipped a **PostgreSQL**-backed session pipeline and a REST API (paginated `/api/sessions`, `/api/export`, `/api/stats`) with API-key middleware, powering a public dashboard with live attacker-activity stats.
- Hardened the honeypot against real adversarial traffic: fixed a concurrency race corrupting session state across parallel SSH channels, bounded connections against flood-triggered OOM, added panic recovery, and closed two post-deployment auth-bypass bugs (unauthenticated `/metrics` endpoint, path traversal in the LLM shell-completion egress gate).

NovaForge | Multi-Factor Approval & Audit Platform | Node.js • TypeScript • Prisma • React Tally CodeBrewers — Finalist

- Built **WebAuthn** passkey login with TOTP fallback and script-hashed recovery codes gated behind a fresh passkey signature; designed trust-decay sessions with immediate per-request revocation in place of hard expiry.
- Engineered an M-of-N / role-based / weighted-quorum approval-policy engine with escalation timeouts and fail-closed fallback, backed by a hash-chained, tamper-evident audit log with non-repudiable WebAuthn-signed votes.

MUJ Placement Kite | Placement Cell Platform | Go • PostgreSQL • Redis • Cloudinary

- Built the student module end-to-end (profiles, draft-request approval workflow, RBAC, rate limiting, Cloudinary document uploads) and an idempotent bulk-upload pipeline that generically parses UG/PG Excel templates.
- Designed audit logging across **12 write paths** and placement-cycle hiring-workflow tracking (round pass/fail, roster + Excel export); built an SMTP worker pool over **Redis Streams** for async mail delivery.

OPEN SOURCE

Honeynet Project | *EventHorizon* | github.com/honeynet/eventhorizon

- Added real-time attacker input capture to the **Telnet tarpit** in **C** and extended the **Go** Prometheus exporter with per-session behavioral metrics; PR merged to main by Honeynet maintainers.

TECHNICAL SKILLS

Languages: Go, Python, Bash, TypeScript, C, Java

Frameworks & Tools: FastAPI, Next.js, Docker, Caddy, Git, GitHub Actions, Ollama, sentence-transformers

DevOps & Infrastructure: Docker Compose, Cloudflare, Tailscale, Prometheus, Grafana, MinIO, Linux

Databases: PostgreSQL, Redis, MongoDB, SQLite, Supabase